

MASS-ENG-010

Notification Engine Specification

Page 1 – Purpose & Scope

Purpose

Define the enterprise notification subsystem responsible for delivering messages, alerts, reminders, and communications across all supported channels.

Objectives

- Centralize outbound communications
- Support email, SMS, push, and in-app notifications
- Respect user preferences and delivery policies
- Provide reliable, auditable delivery

Out of Scope

Business workflow decisions and channel-specific business content.

Page 2 – Architecture

Core Components

Notification Service, Channel Adapters, Template Manager, Preference Manager, Delivery Queue, Retry Manager, Delivery Audit.

Relationships

Consumes Enterprise Core, Identity, Security, Event Bus, Workflow Engine, and AI Orchestration. Receives events and delivers notifications through configured channels.

Page 3 – Functional Specification

Requirements

1. Accept notification requests.
2. Resolve recipient preferences.
3. Select delivery channels.
4. Retry transient delivery failures.
5. Record delivery outcomes.

Non-Functional

Reliable delivery, scalable queues, extensible channel adapters, observability, secure handling of recipient data.

Page 4 – Interfaces & Examples

Interfaces

SendNotification(), QueueNotification(), ResolvePreferences(), TrackDelivery(), RetryDelivery().

Example Flow

Workflow completes → Event Bus publishes CompletionEvent → Notification Engine resolves recipients and preferences → Email and in-app notifications delivered → Delivery audit recorded.

Page 5 – Construction Package

Claude Checklist

Implement notification service; channel adapter framework; template manager; preference service; delivery queue; retry logic; audit logging; integration and unit tests.

Definition of Done

Notifications are reliably delivered according to enterprise policy with configurable channels,

delivery tracking, and complete audit history.

Constitution References

Governance, Security, Identity, Platform Principles, Communication Policies.